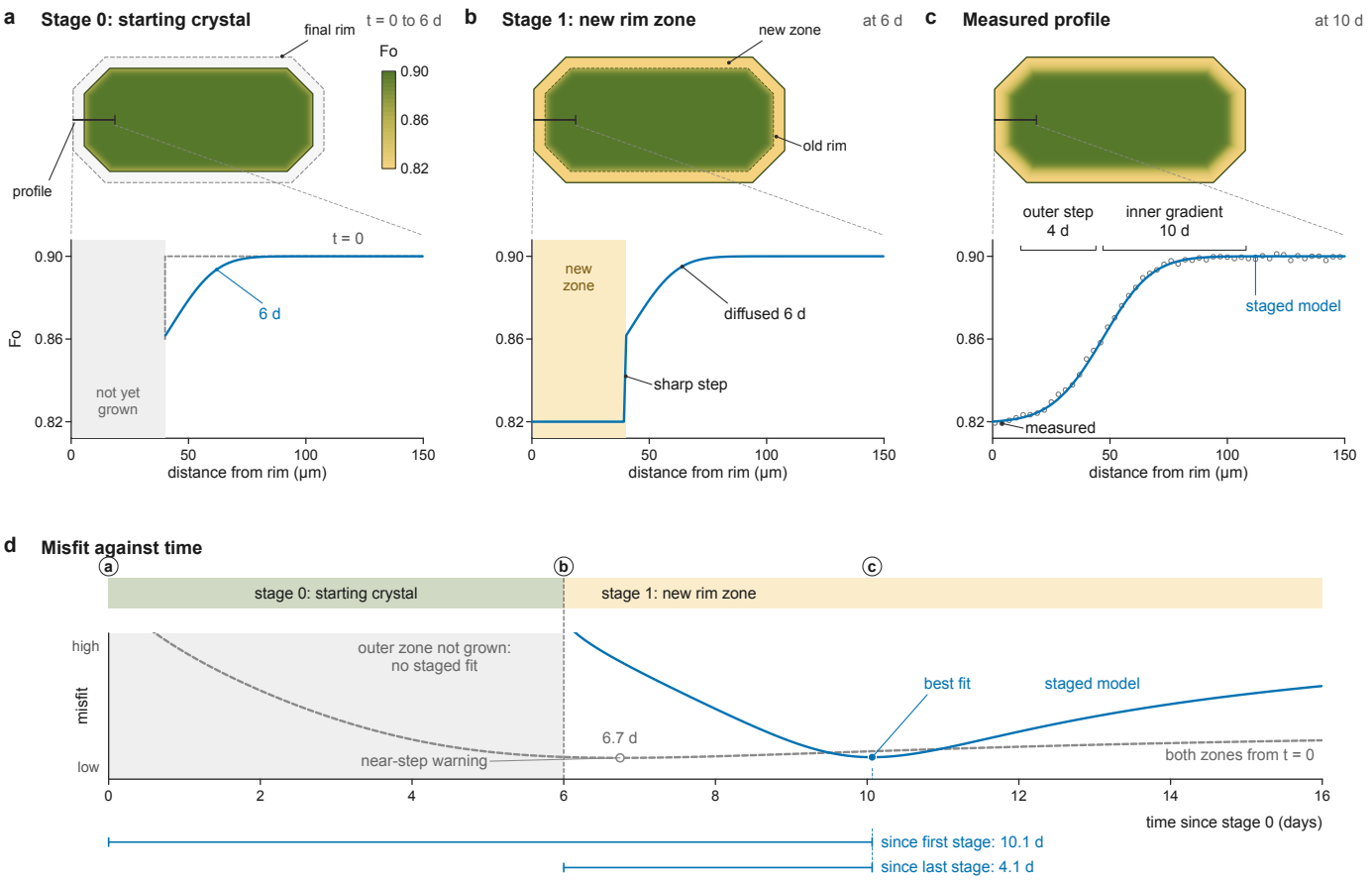

a
Stage 0: starting crystal
t = 0 to 6 d
final rim
Fo
0.90
0.86
0.82
profile
t = 0
6 d
not yet grown
Fo
0.90
0.86
0.82
0
50
100
150
distance from rim (μm)
b
Stage 1: new rim zone
at 6 d
new zone
old rim
new zone
diffused 6 d
sharp step
0.90
0.86
0.82
0
50
100
150
distance from rim (μm)
c
Measured profile
at 10 d
outer step
4 d
inner gradient
10 d
staged model
measured
0.90
0.86
0.82
0
50
100
150
distance from rim (μm)
d
Misfit against time
a
b
c
stage 0: starting crystal
stage 1: new rim zone
high
misfit
low
outer zone not grown:
no staged fit
best fit
staged model
6.7 d
near-step warning
both zones from t = 0
0
2
4
6
8
10
12
14
16
time since stage 0 (days)
since first stage: 10.1 d
since last stage: 4.1 d